

DOMINIQUE BRADSHAW

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EDUCATION

New York University (NYU), New York, NY

Expected Graduation May 2028

- *M.S. in Emerging Technologies (Machine Learning, Artificial Intelligence, and Ethics)* -- Incoming Fall 2026
- Awards: **GEM Fellow**

Washington University in St. Louis, St. Louis, MO

May 2026

- *B.S. in Data Science*, Minor in *African and African American Studies*
Relevant Coursework: Data Structures and Algorithms, Data Science, Matrix Algebra, Artificial Intelligence, Statistics for Data Science, Data Mining, Linear Statistical Models, Machine Learning, Responsible Data Science, Statistical Learning
- Awards: Dean's List (5x), Questbridge Scholar, Chancellor's Career Fellow

TECHNICAL SKILLS

- **Technical Languages:** SQL, Python, R
- **Data & ML:** Pandas, NumPy, scikit-learn, XGBoost, PyTorch, TensorFlow, Fairlearn, LlamaIndex, Jupyter, Git/GitHub, Feature Engineering, Predictive Modeling, Model Evaluation, Exploratory Data Analysis, Data Wrangling, Statistical Analysis, Data Visualization, Business Intelligence
- **BI & Automation:** Power BI, Tableau, Power Automate, Power Automate Desktop, Copilot Studio, SharePoint
- **AI/LLM Skills:** LLM Evaluation & RLHF, Retrieval-Augmented Generation (RAG), Prompt Engineering, Codex, Custom GPTs, Conversational AI, LLM Applications, AI-Assisted Automation, Workflow Automation, Human-in-the-Loop Systems, AI Enablement, AI Tool Integration, Fairness Auditing, Responsible AI Deployment, Bias Mitigation
- **Other:** REST APIs, JSON, Technical Documentation

SELECTED WORK EXPERIENCE

AI Intern, First American

June 2026 – Current

- Sole owner of AI enablement and automation for the Compliance Risk Oversight Support (CROS) team, researching AI tools, building integrations, and driving practical adoption across the department.
- Built 3 Power BI dashboards over SharePoint infrastructure, providing real-time visibility into business continuity, audit remediation, and badge issuance workflows.
- Designed Power Automate workflows automating 100% of follow-up email drafting and routing for project tracking, audit remediation, badge issuance, and facilities risk resolution.
- Built a custom GPT for secure parsing of client questionnaires into structured Excel outputs; authored a prompt library, internal prompt-engineering guide, and a team-specific AI enablement document mapping tools to each CROS role.

AI Trainer, Handshake AI

Jan 2026 – Current

- Produced 100+ weekly RLHF-style evaluations of LLM outputs on data science and data analysis tasks, providing both pairwise preference rankings and detailed written critiques to support model alignment and post-training.
- Authored evaluation rubrics defining optimal output characteristics across reasoning, factuality, code correctness, and analytical depth, surfacing recurring failure patterns to inform iterative model refinement.

AI Intern, Evans and Dixon

Apr 2026 – June 2026

- Designed and implemented AI-assisted workflow automations using Microsoft Power Automate, Power Automate Desktop, Microsoft Copilot Studio, and PowerShell scripting to streamline onboarding and internal operational workflows.
- Engineered automated email, document-processing, and SharePoint-integrated workflows with conditional logic and human-in-the-loop approval systems to reduce repetitive manual tasks and improve process efficiency.
- Authored prompt engineering documentation supporting responsible LLM adoption across organizational teams.

Data Research Assistant, Anthropology Lab, Washington University in St. Louis

Sept 2025 – May 2026

- Managed and verified research data through structured entry and quality checks to support ongoing anthropological studies (**15 hrs/week**)

Teaching Assistant, Washington University in St. Louis

Aug 2024 – May 2026

- Supported 100+ students across Introduction to Data Science and graduate-level Data Mining courses covering Python, statistical analysis, exploratory data analysis, supervised/unsupervised machine learning, ensemble methods, and model evaluation through office hours, grading, and technical instruction.

Resident Advisor, Washington University in St. Louis

Aug 2024 – May 2026

- Served as a resource and mentor for **100+** residents providing guidance on academic, personal, and social matters

Data Science Intern, Meta, Reality Labs Analytics Metaverse Horizon Experiences Team

May 2025 - Aug 2025

- Wrote SQL queries against behavioral data from 6M+ monthly active users to support metric tracking and product experimentation across Horizon Experiences.
- Authored hypothesis-driven analyses on user engagement and driver retention, translating query results into written recommendations for the product team.
- Translated analytical insights into product recommendations, influencing personalization strategies across Horizon Experiences.

External Entity Data Specialist Project Lead, Office of Conflicts of Interest, Washington University in St. Louis

Dec 2023 - June 2025

- Independently launched and led a data integrity initiative, guiding team practices for handling confidential donor and entity data (**20 hrs/week**).

Technology Developer Fellow and EDGE Participant, Sponsors for Educational Opportunity (SEO)

May 2024 - May 2025

- Completed 10+ hours of pre-work in data structures and algorithms followed by a 6-week intensive summer training, contributing to 2 full-stack development projects including an interview prep tool integrating the OpenAI API to generate tailored practice questions for students.

SELECTED PROJECTS & LEADERSHIP EXPERIENCE

Pfizer Extern RAG — Clinical Supply Document Q&A [github.com/mbdominique/pfizer-extern-rag]

- Retrieval-augmented Q&A system built during Extern's AI-Powered Document Insights externship with Pfizer. Full pipeline: PyMuPDF extraction, sentence-based chunking, hybrid retrieval combining dense embeddings and BM25 with Reciprocal Rank Fusion, cross-encoder reranking, and grounded generation with Gemini. Includes labeled eval harness; achieves 86% top-3 retrieval accuracy.

Fairness Interventions in Medical Prediction Models

- Audited demographic bias in a hospital readmission model on the diabetes hospital dataset. Trained a baseline Random Forest, then implemented pre-processing (CorrelationRemover), in-processing (Adversarial Fairness Classifier), and post-processing (ThresholdOptimizer) interventions using Fairlearn. Compared accuracy-fairness tradeoffs across 10 random seeds using demographic parity, equalized odds, and selection rate metrics.

Customer Transaction Prediction Project, Data Analysis, Machine Learning, Exploratory Analysis

- Built an **XGBoost** classification model to predict customer transactions with an **accuracy of 90%** and **AUC of 0.85** on the validation set.
- Conducted **exploratory data analysis (EDA)** on a dataset of 200,000 records (200+ features) to assess distributions and address 90:10 class imbalance via **class weighting** using **scikit-learn** and **pandas**.

Management Leaders of Tomorrow (MLT) Ascend Scholar | National Society of Black Engineers, Programs Zone Chair | Community Engagement Chair, Delta Sigma Theta Sorority Incorporated

JP Morgan Chase Data for Good Hackathon, Data Science, Machine Learning

- Collaborated with a team to build an innovative solution to address a real-world data challenge given by a non-profit organization
- Utilized Python, Pandas, and Scikit-Learn to analyze datasets and build a **Random Forest** model to predict areas most in need of a non-profits solutions.